EXPLORING RISK IN REAL-WORLD AI SYSTEMS

CAUSAL TAXONOMY ANALYSIS

After reviewing the MIT AI Risk Scenario Explorer's Causal Taxonomy, I selected the following categories as most relevant to my professional context in financial services AI systems, particularly fraud detection platforms.

Entity: Organization/Deployer
Intent: Negligence
Timing: Deployment

Entity: Organization/Deployer
In my context, the deploying organization is ultimately responsible for AI outcomes because it controls model integration, validation, governance, and monitoring. In a fraud detection system, even if models are developed by vendors or internal teams, the organization decides how they are deployed into production. If governance is weak or validation is insufficient across customer segments, the system may produce biased or unreliable fraud decisions that directly impact customers.

Intent: Negligence
The most realistic risk source is negligence rather than malicious intent. This can occur when teams rush deployment, underinvest in testing, or fail to validate data quality. For example, a fraud model may not be properly tested for edge cases such as seasonal shopping spikes or international transactions, resulting in legitimate payments being incorrectly blocked and creating financial and reputational harm.

Timing: Deployment
Deployment is where risks become real because models interact with live data and evolving fraud patterns. Even well-trained models can degrade once exposed to production environments. Fraud tactics change rapidly, and without monitoring, model drift may reduce accuracy and allow fraudulent transactions to pass undetected.

Overall, this taxonomy highlights that AI risk is driven more by organizational decisions and operational discipline than by model design alone.

DOMAIN TAXONOMY OF AI RISKS

The three most relevant AI risk domains in financial services fraud detection are **Privacy & Security**, **Human-Computer Interaction**, and **AI System Safety, Failures, and Limitations**.

These domains are particularly important because fraud detection systems operate on large volumes of sensitive financial data and support decisions that directly affect customers.

Privacy & Security is a critical concern because fraud detection platforms process personal and financial information, making them attractive targets for cyberattacks. Potential risks include data breaches, unauthorized access, data leakage, and inference attacks on model outputs. To mitigate these threats, organizations should implement strong security controls such as encryption, role-based access control, data masking, zero-trust architecture, and continuous security monitoring. These measures reduce the likelihood

and impact of security incidents while supporting compliance with financial regulations and privacy requirements.

Human-Computer Interaction risks arise when analysts place excessive trust in AI-generated recommendations. For example, a legitimate customer may be incorrectly flagged because of unusual spending behavior, and analysts who rely solely on AI outputs may block valid transactions unnecessarily. Effective mitigation strategies include explainable AI, confidence scoring, user training, and human-in-the-loop review processes. These approaches improve transparency and help ensure that critical decisions are validated by human judgment.

AI System Safety, Failures, and Limitations become significant when models degrade over time due to changing fraud patterns, poor data quality, or bias in training datasets. This can increase false positives and false negatives, reducing system effectiveness. Organizations should implement continuous performance monitoring, bias testing, periodic retraining, and independent model validation before deployment updates.

Together, these safeguards create a more trustworthy and resilient AI system. When combined with strong governance and human oversight, they help ensure that fraud detection models remain reliable, accurate, and effective in dynamic real-world environments.

WORKPLACE REFLECTION

A relevant example from my professional experience is an **AI-powered fraud detection system** used in the financial services industry. The system processes millions of transactions in real time and uses machine learning models to assign risk scores that identify potentially fraudulent activity. By automating fraud detection, the platform improves operational efficiency, reduces manual review effort, enhances customer protection, and enables faster responses to suspicious transactions.

Key Risks

- **Privacy & Security:** Fraud detection systems rely on large volumes of sensitive customer and financial data, increasing the risk of cyberattacks, unauthorized access, data breaches, and regulatory compliance violations.
- **Human-Computer Interaction:** Analysts may develop automation bias and place excessive trust in AI-generated risk scores, potentially overlooking contextual information that could improve decision quality.
- **AI System Safety, Failures, and Limitations:** Fraud tactics evolve continuously, which can cause model drift and reduce accuracy over time, leading to false positives, customer friction, and missed fraudulent transactions.

Risk Mitigation Strategies

- Implement strong data governance practices including Encryption of sensitive data, Role-based access controls, Audit logging and continuous security monitoring

- Use **Explainable AI (XAI)** to provide transparency into model decisions and help analysts understand why transactions are flagged.
- Maintain **human-in-the-loop review** for high-risk transactions and critical decision-making processes.
- Conduct continuous performance monitoring, bias testing, model validation, and scheduled retraining to maintain reliability and fairness.

Reflection

Similar to healthcare AI systems such as PathAI, fraud detection AI is most effective when it supports and augments human expertise rather than replacing it entirely. Responsible AI requires strong governance, transparency, accountability, and ongoing oversight to ensure trustworthy, ethical, and effective outcomes for both customers and financial institutions.

Conclusion

This exercise reinforced that successful AI deployment depends as much on governance and oversight as on model accuracy. In my role, I would prioritize continuous monitoring and human review processes to ensure fraud detection systems remain fair, reliable, and aligned with business objectives.